NAJJAR Nour Elhouda

Data Engineer | Data Science & AI Engineer

Tunis, Tunisia | +216 52 235 320 | nourelhouda.najjar@esprit.tn

LinkedIn: https://www.linkedin.com/in/nour-el-houda-najjar/ | GitHub: https://github.com/nournajjjar

Portfolio: https://nournajjjar.github.io

PROFESSIONAL SUMMARY

Data Engineer and Computer Science Engineer (ESPRIT, engineering degree, Data Science specialization) with hands-on experience designing **data pipelines**, ETL/ELT workflows, and production AI systems. Over 9 months of industry experience at VALUE Digital Services building a containerized AI platform for corporate credit analysis: PDF data extraction, PostgreSQL/pgvector data modeling, LLM and RAG services, stress testing, and forecasting engines, all monitored with MLflow. Strong foundation in **Data Engineering** (Apache Airflow, Spark, Hadoop, SQL, NoSQL), programming (Python, Java, C++), and Machine Learning. Known for interpersonal skills, creativity, and turning complex, data-heavy problems into reliable, measurable, deployed results (92% churn-prediction accuracy, sub-500 ms analytics engines, 1.0 chatbot faithfulness).

CORE COMPETENCIES

Data Engineering · Data Pipelines · ETL / ELT · Data Modeling · Data Integration · Big Data (Hadoop, Spark) · Workflow Orchestration (Apache Airflow) · SQL & Database Design · Machine Learning · MLOps · LLMs & RAG · Cloud (Azure) · Docker & Containerization · REST APIs · Data Quality & Monitoring

PROFESSIONAL EXPERIENCE

AI & Data Engineer – End-of-Studies Internship **12/2025 – 06/2026**

VALUE Digital Services *Tunis, Tunisia*

Project: "AI-Driven Agentic Framework for Corporate Banking" – full-stack platform automating credit analysis for Tunisian industrial companies.

- Designed and deployed an end-to-end AI and data platform for corporate credit analysis, integrating LLMs, retrieval-augmented generation (RAG), financial stress testing, and economic forecasting into a production-ready system.
- Built a **data extraction pipeline** for financial statements in PDF (PyMuPDF, Docling, Phi-4 via Azure OpenAI), transforming unstructured documents into structured, validated data stored in PostgreSQL, reducing manual data entry.
- Designed the data model and vector store (PostgreSQL + pgvector, multilingual MiniLM embeddings) powering a RAG Q&A chatbot based on cosine-similarity retrieval, reaching a **faithfulness score of 1.0**.
- Engineered a **16-section automated credit report generator** with built-in quality scoring, a 6-attempt retry loop, and LLM-as-Judge evaluation, improving output reliability and consistency.
- Developed a stress testing engine (3 macroeconomic scenarios + custom scenarios) and a bisection-based reverse stress test identifying the breaking point at a **∼59% revenue drop**; runs in **413 ms**.
- Built an OLS revenue forecasting module with confidence intervals and scenario planning, running in **482 ms**.
- Integrated a web research data source (Tavily API) covering ∼12 themes, with a post-fetch relevance filter preventing cross-company data pollution.
- Refactored a monolithic script into a modular FastAPI architecture with asynchronous background jobs, step-by-step progress tracking, and JWT authentication.
- Containerized and deployed the full stack (FastAPI, React, PostgreSQL, pgvector, Docker Compose) with experiment tracking and LLM observability via MLflow and Laminar; defined a KPI framework across 6 platform modules.

AI & Data Science Intern **06/2025 – 08/2025**

VALUE Digital Services *Tunis, Tunisia*

- Built a multi-agent simulation of **200+ virtual clients** using Mesa and Monte Carlo methods to model customer churn and economic scenarios.
- Developed a RAG system (FAISS + LLaMA) for automated client-profile generation and behavioral simulation.
- Prepared, cleaned, and engineered features on simulated customer data to train churn models, achieving **92% accuracy** in churn prediction and supporting data-driven retention decisions.

Technical Intern **07/2024 – 08/2024**
CERT – Telecommunications Research and Studies Center *Tunis, Tunisia*

- Conducted frequency and equipment testing across GSM, LTE, Wi-Fi, and Bluetooth technologies.
- Performed technical verification against IEEE 802.15 standards and documented measurement results in structured test reports.

ACADEMIC PROJECTS

AI Platform for Crypto Due Diligence and Forecasting
Python, MySQL, Apache Airflow, BERT, LLaMA 3, GPT-2, Flask

- Designed automated **ETL data pipelines** orchestrated with Apache Airflow, aggregating on-chain, market, and news data into a MySQL database.
- Integrated BERT, LLaMA 3, and GPT-2 for classification and insight generation, plus a RAG Q&A system over crypto whitepapers.
- Delivered an interactive Flask dashboard for real-time exploration of forecasts and insights.

Smart Greenhouse Management Platform (IoT + AI)
FastAPI, PostgreSQL, MQTT (Mosquitto), React, TypeScript, ESP32, Arduino, scikit-learn, YOLOv8

- Designed an IoT data pipeline collecting real-time sensor data (pH, NPK, moisture) from Arduino/ESP32 devices (UART, WiFi) into a FastAPI + PostgreSQL backend, with MQTT actuator commands and WebSocket live dashboards.
- Implemented multi-tenant data modeling (users, greenhouses, towers), an automation rules engine, and stale-sensor detection for data reliability.
- Trained a Random Forest yield regressor ($\mathbf{R^2 = 0.844}$, MAE = 3.81 g) and a 5-phase germination classifier; built a YOLOv8 leaf-defect detection pipeline.

Auto-Prompting RAG Pipeline
Python, OpenAI / Groq / OpenRouter APIs, Streamlit

- Built a modular RAG pipeline (image extraction, prompt generation, evaluation) with a self-improving loop driven by "Prompt Researcher" and "Prompt Engineer" agents, deployed with Streamlit.

EDUCATION

Engineering Degree in Computer Science (Master's level, Bac+5) – Data Science Specialization **09/2023 – 06/2026**
ESPRIT – Private Higher School of Engineering and Technology *Tunis, Tunisia*
Relevant coursework: Data Engineering, Big Data (Hadoop, Spark), Databases & SQL, Data Warehousing, Machine Learning, Deep Learning, NLP, Cloud Computing, Software Engineering.

Preparatory Program in Mathematics and Physics (Bac+2) **09/2020 – 06/2023**
Faculty of Sciences of Monastir *Monastir, Tunisia*

TECHNICAL SKILLS

- **Programming Languages:** Python (NumPy, Pandas, Scikit-Learn, TensorFlow), SQL, Java, C, C++, C#, JavaScript, TypeScript
- **Data Engineering:** ETL/ELT, Data Pipelines, Apache Airflow, Apache Spark, Hadoop, Data Modeling, Data Integration, Web Scraping & APIs
- **Databases:** PostgreSQL, pgvector, MySQL, NoSQL, Azure Cosmos DB, FAISS (vector search)

- **Machine Learning & AI:** Machine Learning, Deep Learning, NLP, LLMs, RAG, Prompt Engineering, LLM-as-Judge, YOLOv8, Forecasting (OLS, Holt), Financial Stress Testing, Monte Carlo Simulation, Mesa
- **Cloud, DevOps & MLOps:** Microsoft Azure (Azure OpenAI), Docker, Docker Compose, MLflow, Laminar, Langfuse, Git, CI/CD basics
- **Backend & Frameworks:** FastAPI, Flask, Spring Boot, .NET, Node.js, React, REST APIs, MQTT, WebSocket, JWT
- **Visualization & BI:** Power BI, Streamlit, Matplotlib, Recharts
- **Methodologies:** CRISP-DM, TDSP, Agile/Scrum, Design Thinking

CERTIFICATIONS

- Building Transformer-Based NLP Applications – NVIDIA Deep Learning Institute, 10/2025
- AI Applications for Anomaly Detection – NVIDIA Deep Learning Institute, 05/2025
- Hashgraph Developer Certification – Hedera, 11/2025

SOFT SKILLS

Interpersonal skills · Creativity · Teamwork & collaboration · Communication (technical and non-technical audiences) · Problem solving · Autonomy & adaptability

VOLUNTEERING & LEADERSHIP

Member
Aerospace Kairouan *Kairouan, Tunisia*

- Took part in the design and construction of aircraft prototypes as part of a team, from concept to physical build.
- Strengthened teamwork, creativity, and hands-on engineering problem-solving skills.

Member – Sponsorship Department
JCI Kairouan (Junior Chamber International) – 2 years *Kairouan, Tunisia*

- Member of the Sponsorship Department, contributing to sponsor relations and fundraising for the chapter's events.
- Co-organized a school supplies donation campaign: collected supplies and distributed them to students and people in need.
- Co-organized a Ramadan food donation campaign, collecting food and delivering it to families in need.
- Contributed to organizing music events and a marathon.

Active Member
Enactus – Faculty of Sciences of Monastir *Monastir, Tunisia*

- Contributed to organizing musical and educational events, strengthening teamwork and interpersonal skills.

LANGUAGES

Arabic: Native | French: Professional working proficiency (B2) | English: Professional working proficiency (B2)